But we won’t delve into NN-search for R-trees: now that readers should have a high-level understanding of this data structure, we are ready to move to its evolution, the SS-tree.

It’s also worth mentioning, as a final gloss, that R-trees do not guarantee good worst-case performance, but in practice they usually perform better than k-d trees, and so they were, for a long time, the de-facto standard for similarity search and indexing of multidimensional datasets.

10.3 Similarity Search Tree

We have seen, in section 10.2, some of the key properties that influence the shape and performance of R-trees; let’s recap them here:

1. The splitting heuristic;
2. The criteria used to choose the sub-tree to add new points (if more than one overlaps);
3. The distance metric.

For R-trees, we assumed that aligned boxes, hyper-rectangles parallel to the Cartesian axes, are used as bounding envelopes for the nodes. If we lift this constraint, the shape of the bounding envelope becomes the 4th property of a more general class of similarity search trees.

And, indeed, to its core, the main difference between R-trees and SS-trees, at least in their most basic version, is exactly the shape of bounding envelopes: as shown in figure 10.9,